Multigrid Finite State Projection for the Reaction-Diffusion Master Equation

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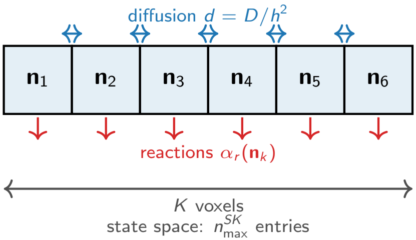
The RDME: reactions and diffusion across a spatial grid

Model. Divide a 1-D domain into K voxels. Each voxel k holds molecule counts $\mathbf{n}_k \in \mathbb{Z}^S_{\geq 0}$.

Two types of events:

- **Reactions** inside voxel k: rate $\alpha_r(\mathbf{n}_k)$, change $\mathbf{n}_k$.
- **Diffusion** between neighbors: rate $d = D/h^2$, moves one molecule $k \leftrightarrow k+1$.

The joint probability $p(\mathbf{n}_1, \dots, \mathbf{n}_K, t)$ satisfies $\dot{p} = Ap$ **(RDME)**.



The computational bottleneck

The state space has $\sim n_{\max}^{SK}$ reachable states — **exponential in** K. Even $K{=}8$, $S{=}1$ is intractable with a naive FSP.

$\dot{p} = Ap$ **has block-tridiagonal structure** (schematic, K=4, S=1)

$$\begin{array}{cc} & \begin{array}{cccc} \mathbf{n}_1 & \mathbf{n}_2 & \mathbf{n}_3 & \mathbf{n}_4 \end{array} \\ \begin{array}{c} \mathbf{n}_1 \\ \mathbf{n}_2 \\ \mathbf{n}_3 \\ \mathbf{n}_4 \end{array} & \begin{bmatrix} A_1 & D_{12} & & \\ D_{12} & A_2 & D_{23} & \\ & D_{23} & A_3 & D_{34} \\ & & D_{34} & A_4 \end{bmatrix} \end{array} \cdot \begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} \dot{p}_1 \\ \dot{p}_2 \\ \dot{p}_3 \\ \dot{p}_4 \end{bmatrix}$$

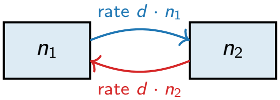
each block: $n_{\max}^{S} \times n_{\max}^{S}$ total: $n_{\max}^{SK} \times n_{\max}^{SK}$

A_k: reactions inside voxel k $D_{k,k+1}$: diffusion between neighboring voxels p_k: probability sub-vector for voxel k

The Binomial is the exact diffusion equilibrium

Fix total molecules $\bar{n} = n_1 + n_2$ across two adjacent voxels.

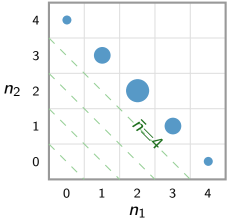
Diffusion moves one molecule at a time, at rate d per molecule:



Each molecule hops independently $\Rightarrow$ at equilibrium, equally likely to be in either voxel. For $\bar{n}$ total:

$$\boxed{P(n_1 \mid \bar{n}) = \mathrm{Bin}\big(\bar{n}, \tfrac{1}{2}\big)} \quad (\text{exact, any } d > 0)$$

When the distribution is near this equilibrium, $\bar{n}$ **alone describes the two-voxel state**.



Probability spreads along constant-total diagonals as $\mathrm{Bin}(\bar{n}, \frac{1}{2})$ (dot size $\propto$ mass).

The two-voxel state is captured by $\bar{n}$ alone.

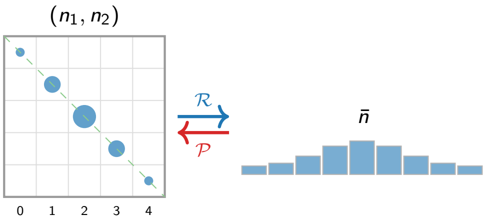
Merging two adjacent voxels: restriction and prolongation

Compress ($\mathcal{R}$) — exact:

$$p^c(\bar{n}) = \sum_{n_1+n_2=\bar{n}} p(n_1, n_2).$$

Reconstruct ($\mathcal{P}$) — approximate:

$$p(n_1, n_2) \approx p^c(\bar{n}) \cdot \mathrm{Bin}(\bar{n}, \tfrac{1}{2}).$$



Summing along constant-total diagonals gives the scalar $\bar{n}$. Reconstruction spreads mass back via Binomial.

State-space reduction

Per pair: $n_{\max}^{2S} \to n_{\max}^{S}$.
All K voxels: $n_{\max}^{SK} \to n_{\max}^{SK/2}$.

$\mathcal{P}$ is accurate when the distribution is near-Binomial — verified at runtime, not assumed globally.

What merging does to the rate matrix

Full matrix A — 4 voxels:

$$\begin{array}{c|cccc|} & n_1 & \text{pair 1} \; n_2 & n_3 & \text{pair 2} \; n_4 \\ \hline n_1 & A_1 & D_{12} & & \\ n_2 & D_{12} & A_2 & D_{23} & \\ n_3 & & D_{23} & A_3 & D_{34} \\ n_4 & & & D_{34} & A_4 \\ \hline \end{array}$$

Size: $n_{\max}^4$

$$\xrightarrow{\mathcal{R}} \qquad \xleftarrow{\mathcal{P}}$$

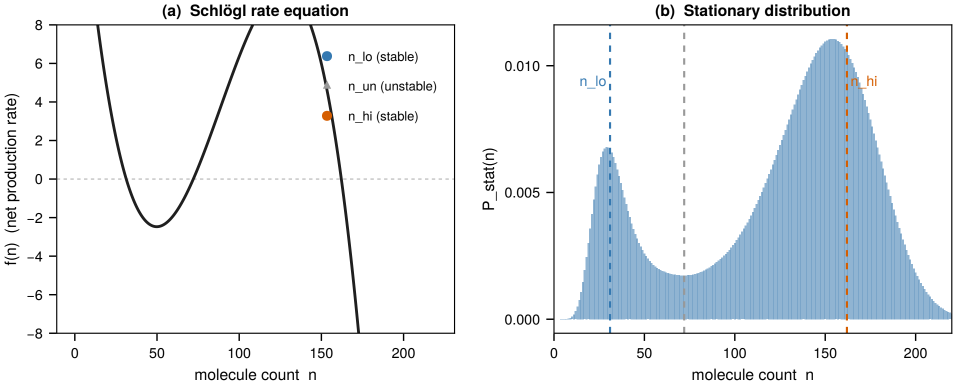
Merged matrix $\bar{A}$ — 2 pairs:

$$\begin{array}{c|cc|} & \bar{n}_1 & \bar{n}_2 \\ \hline \bar{n}_1 & \bar{A}_1 & \bar{D}_{12} \\ \bar{n}_2 & \bar{D}_{12} & \bar{A}_2 \\ \hline \end{array}$$

Size: $n_{\max}^2$ (**10^4**$\times$ smaller for $n_{\max}$=100)

- $A_k \to \bar{A}_j$: reactions averaged over Binomial
- Within-pair D disappears — implicit in Binomial
- $\bar{D}_{12}$: only between-pair diffusion remains

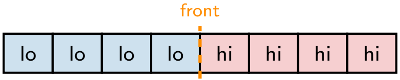
Each voxel is bistable: two stable molecular states



(a) Net production rate $f(n)$ — three zeros: two stable (n_{lo}, n_{hi}), one unstable. **(b)** Stationary distribution is bimodal.

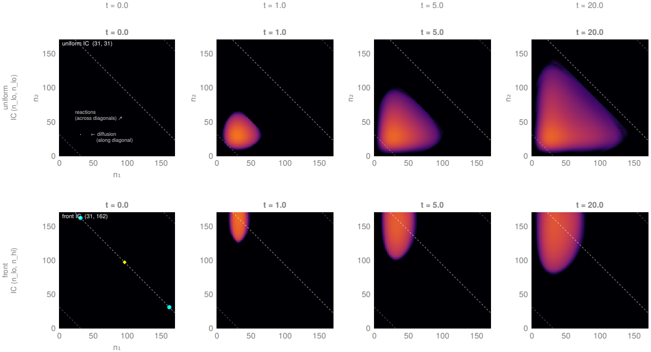
Think of it as a double-well potential: each voxel wants to be at n_{lo} or n_{hi}. Each spatial configuration — e.g., $(n_{\mathrm{lo}}, n_{\mathrm{lo}}, n_{\mathrm{hi}}, n_{\mathrm{hi}}, \ldots)$ — is metastable.

A **spatial front** forms when neighboring voxels occupy opposite stable states. Slow diffusion ($d \ll$ reaction rates) sustains it indefinitely.



At the front: the joint distribution of the two interface voxels is *bimodal* — concentrated at $(n_{\mathrm{lo}}, n_{\mathrm{hi}})$ and $(n_{\mathrm{hi}}, n_{\mathrm{lo}})$.

The Binomial is **unimodal**: wrong shape, wrong means.



Top: uniform IC — probability spreads along constant-total diagonals; Binomial reconstruction is accurate. ✓

Bottom: bistable front — probability stays at the corners; Binomial reconstruction gives the wrong shape. ✗

When is merging valid? The imbalance criterion

Compare mean counts in adjacent voxels $2j-1$ and $2j$:

$$\alpha_j = \frac{|\mu_{2j-1} - \mu_{2j}|}{\mu_{2j-1} + \mu_{2j}}.$$

- $\alpha_j \approx 0$: voxels balanced $\Rightarrow$ distribution near-Binomial $\Rightarrow$ **safe to merge**.
- $\alpha_j \approx 1$: voxels in opposite states $\Rightarrow$ front is nearby $\Rightarrow$ **keep fine**.

Adaptive partitioning

Find $j^* = \text{argmax}_j\, \alpha_j$ (front location).
Keep j^* and its immediate neighbors fine.
Merge all other pairs.

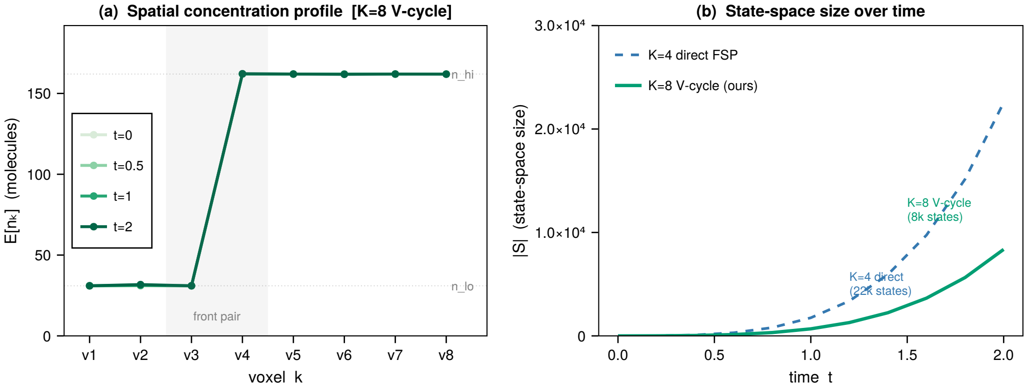
Recheck α_j only when the front moves.

One V-cycle step:

1. **Detect front** via α_j.
2. **Restrict** bulk pairs: $\bar{n}_j = n_{2j-1} + n_{2j}$.
3. **Coarse solve**: $\dot{p}^M = A^M p^M$ on $K/2$ effective voxels.
4. **Prolong** fine pairs back using dynamic-π (correct at front; Binomial elsewhere).

State space: $K/2$ coarse voxels + small fine window. Tractable where full K-voxel FSP is not.

Results: the V-cycle tracks the front and stays tractable



(a) Spatial concentration profile $\mathbb{E}[n_k]$ vs. voxel k at four times — K=8 Schlögl, V-cycle solver. The step-function front (voxels 1–3 at n_{lo}, voxels 4–8 at n_{hi}) is tracked exactly throughout.

(b) State-space size over time. K=8 V-cycle uses *fewer* states than K=4 direct FSP, while solving a system twice as large. Direct K=8 FSP is intractable.

Summary

1. Fast diffusion enables state-space compression.

When the joint distribution of two adjacent voxels is near $\mathrm{Bin}(\bar{n}, \frac{1}{2})$, replacing (n_1, n_2) by $\bar{n}$ cuts the per-pair state space from $n_{\max}^{2S}$ to $n_{\max}^{S}$.

2. Bistable fronts violate the Binomial assumption.

Reactions keep neighboring voxels in opposite stable states, producing a bimodal conditional distribution. The imbalance α_j detects this automatically at runtime.

3. Adaptive V-cycle FSP.

Keep a fine window at the front; coarsen everything else. Evolve $\dot{p}^M = A^M p^M$ directly. **Accurate and tractable where full FSP is not.**

K=8 V-cycle: 8,361 states vs K=4 direct FSP: 22,551 states. Direct K=8 FSP: intractable.

Open: rigorous error bounds; multi-species; multiple/moving fronts.

Appendix: why not reconstruct at every step?

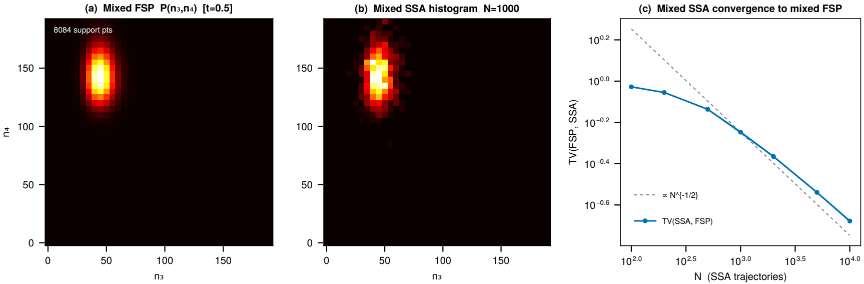
$$
\begin{tikzcd}
\boxed{p \quad \text{(fine)}} \arrow[d, "\text{pre-smooth}"] \\
\boxed{\tilde{p} \quad \text{(smoothed)}} \arrow[d, "\mathcal{R}\text{: compress}"] \\
\boxed{p^c \quad \text{(coarse)}} \arrow[d, "e^{\Delta t \bar{A}}"] \\
\boxed{\tilde{p}^c \quad \text{(advanced)}} \arrow[d, "\mathcal{P}\text{: reconstruct}"] \\
\boxed{p_{\text{new}} \quad \text{(fine)}}
\end{tikzcd}
$$

Reconstruction is wrong at the front

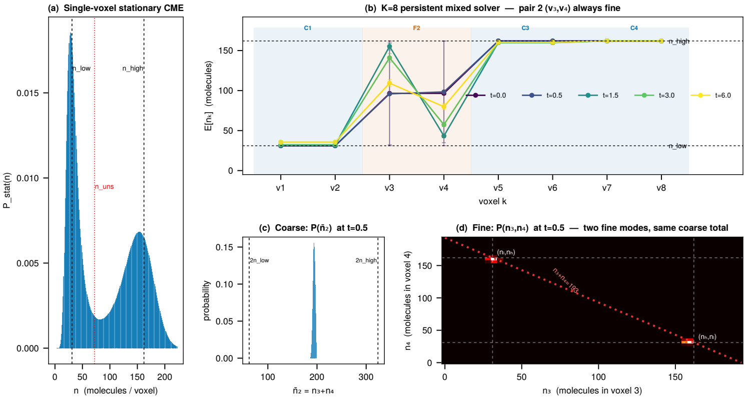
Step 4 forces a Binomial split on *every* pair, including the front where the true distribution is bimodal.

Fix: evolve p^M directly — no reconstruction step. Coarse pairs carry the Binomial assumption implicitly in $\bar{A}_j$; the fine window tracks the true within-pair distribution exactly.

Appendix: mixed-resolution generator is self-consistent



ODE solve on the mixed state $(\bar{n}_1, n_3, n_4)$ vs. Monte Carlo on the same reduced generator. Means agree to $<1\%$; TV $\sim N^{-1/2}$ (sampling noise only).


(a) Single-voxel stationary CME
P_stat(n)
n_low
n_high
n_uns
n (molecules / voxel)
(b) K=8 persistent mixed solver — pair 2 (v₃,v₄) always fine
C1
F2
C3
C4
n_high
n_low
t=0.0
t=0.5
t=1.5
t=3.0
t=6.0
E[nₖ] (molecules)
voxel k
(c) Coarse: P(ñ₂) at t=0.5
2n_low
2n_high
probability
ñ₂ = n₃+n₄
(d) Fine: P(n₃,n₄) at t=0.5 — two fine modes, same coarse total
n₃+n₄=193
n₄ (molecules in voxel 4)
n₃ (molecules in voxel 3)